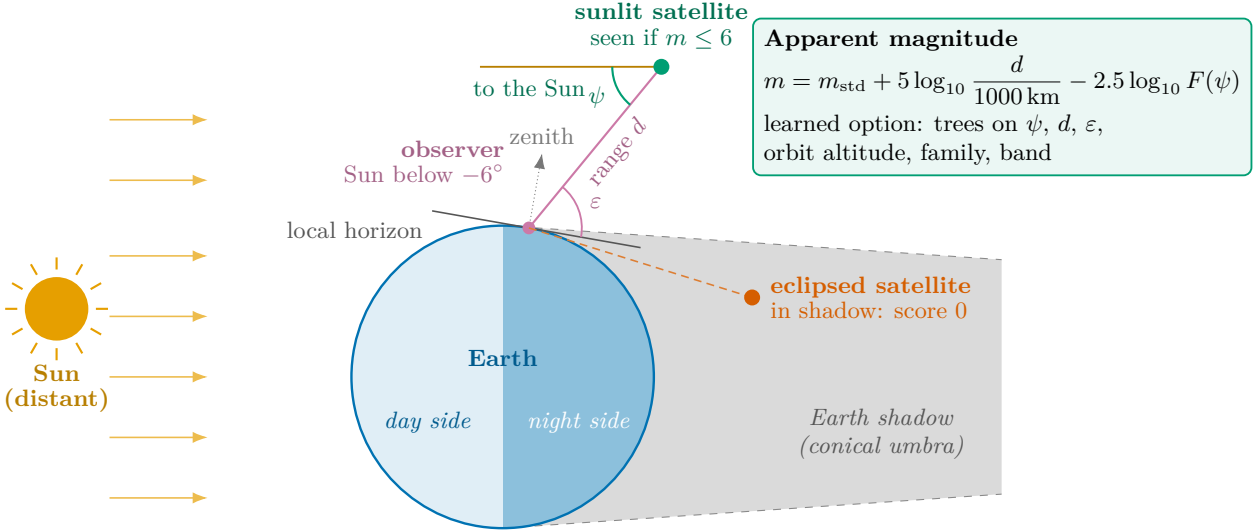
sunlit satellite
seen if $m \le 6$
to the Sun ψ
observer
Sun below $-6°$
zenith
range d
ε
local horizon
eclipsed satellite
in shadow: score 0
Earth
day side
night side
Earth shadow
(conical umbra)
Sun
(distant)
Apparent magnitude
$m = m_{\mathrm{std}} + 5\log_{10}\dfrac{d}{1000\,\mathrm{km}} - 2.5\log_{10} F(\psi)$
learned option: trees on ψ, d, ε,
orbit altitude, family, band